

Transcription Initiation in Prokaryotes

References:

- **Molecular Biology of the Gene, 5th Edition, James D. Watson**
- **Molecular Biology, 5th Edition, Robert F. Weaver**

Until 1980, it was a common assumption that transcription initiation ended when RNA polymerase formed the first phosphodiester bond, joining the first two nucleotides in the growing RNA chain. Then, **Carpousis** and **Gralla** reported that initiation is more complex than that.

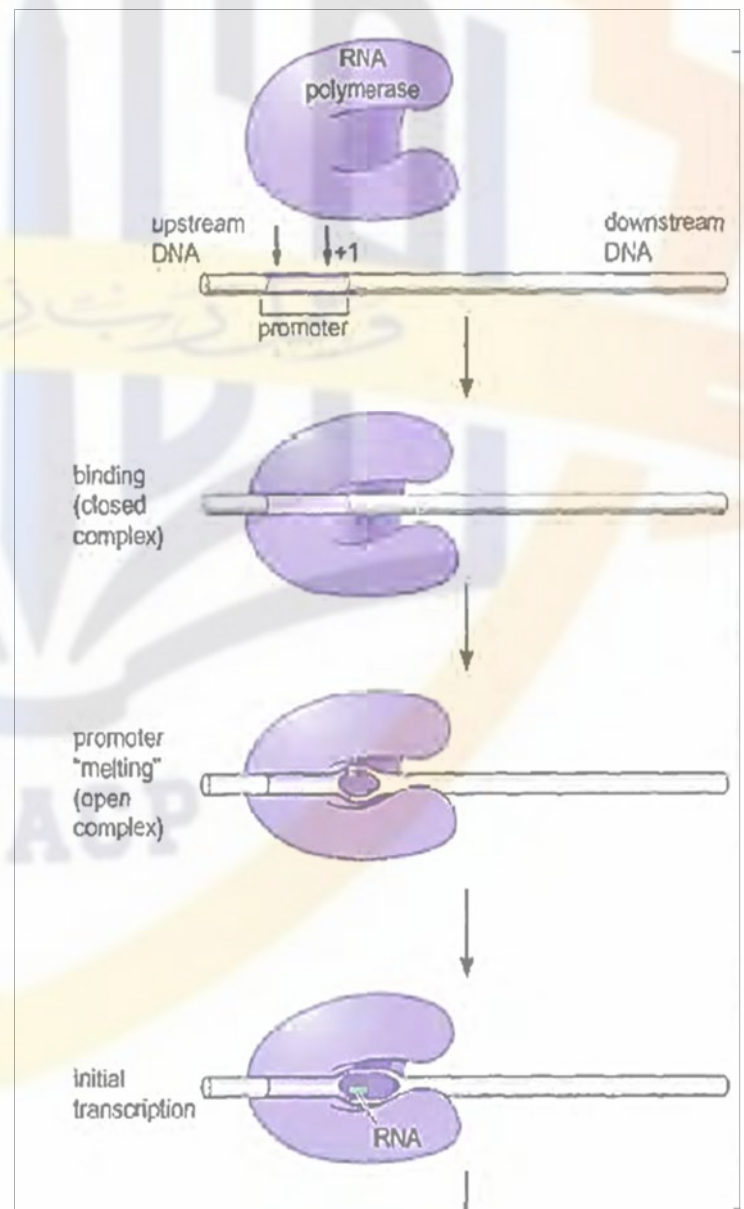
Transcription Initiation Involves Three Defined Steps:

1. Closed Promoter Complex:

- The **first step** is the **initial binding of polymerase to a promoter** to form what is called a **closed complex**.
- In this form the DNA remains double stranded, and the enzyme is bound to one face of the helix.

2. Open Complex:

- In the **second step of initiation**, the closed complex undergoes a transition to the **open complex** in which the DNA strands separate over a distance of some 14 bp around the start site to form the transcription bubble.
- The opening up of the DNA frees the template strand.
- The first two ribonucleotides are brought into the active site, aligned on the template strand, and joined together.

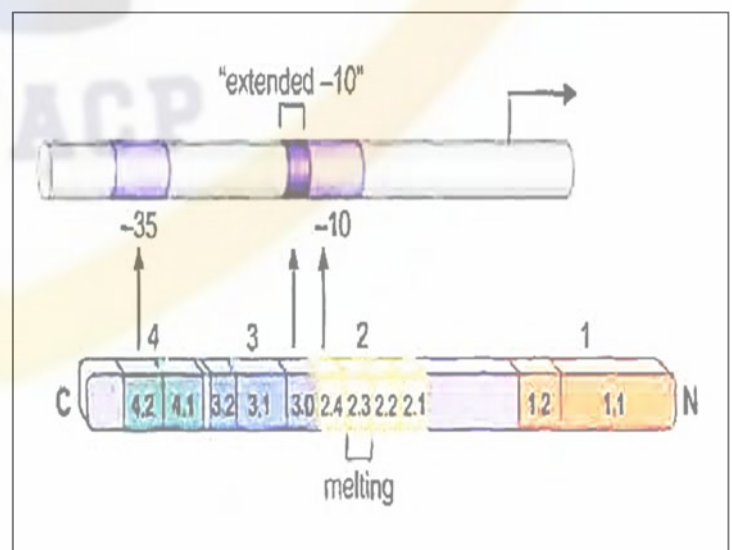


3. Stable Ternary Complex:

- The enzyme then begins to move along the template strand, opening the DNA helix ahead of the site of polymerization and allowing it to resealed behind.
- In this way, subsequent ribonucleotides are incorporated into the growing RNA chain.
- Incorporation of the first ten or so ribonucleotides is a rather inefficient process, and at that stage the enzyme often releases short transcripts (each of less than ten or so nucleotides) and then begins synthesis again.
- Once an enzyme gets further than the 10 bp. it is said to have escaped the promoter.
- At this point it has formed a **stable ternary complex, containing enzyme, DNA, and RNA**. This is the **transition to the elongation phase**.

Step 1) Closed Promoter Complex and Stimulation of Transcription Initiation by Sigma:

- **RNA polymerase binds to DNA in a closed promoter complex.**
- The bacterial core RNA polymerase can, in principle, initiate transcription at any point on a DNA molecule. In cells, however, polymerase initiates transcription only at promoters.
- It is the addition of an **initiation factor called σ** that converts **core enzyme into the form that initiates only at promoters**. That form of the enzyme is called the **RNA polymerase holoenzyme**.
- σ directs tight binding of RNA polymerase to promoters, it places the enzyme in a position to initiate transcription—at the beginning of a gene. **Travers and Burgess** experimentally verified that **σ stimulate initiation of transcription**.
- In the case of **E. coli**, the predominant σ factor is called **σ^{70}** .
- The **σ^{70}** factor can be divided into **four regions** called a **region 1 through a region 4**.
- Region 2 recognizes the -10 element of the promoter
- Region 4 recognizes -35 element of the promoter

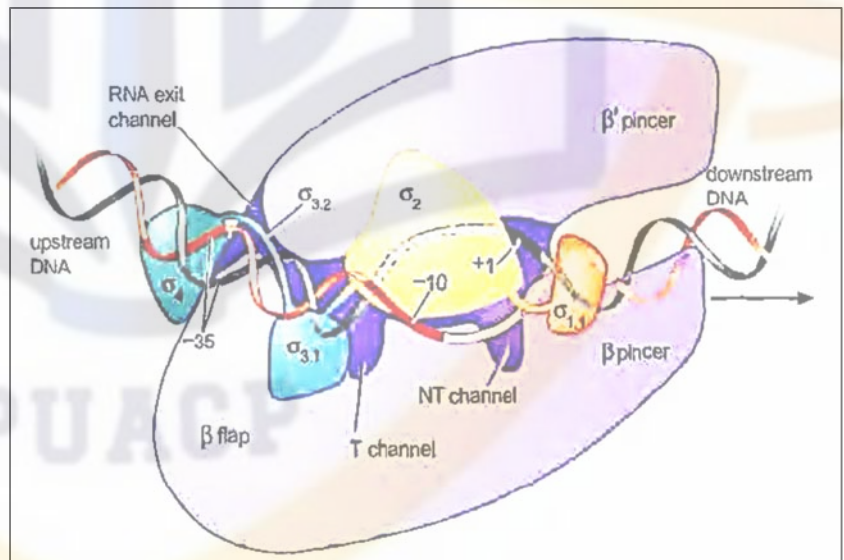


- **Two helices within region 4** form a common DNA-binding motif called a **helix-turn-helix**.
 - **One helix** inserts into the major groove and **interacts with bases in -35 region**;
 - **Other helix** lies across the top of the groove, making **contacts with the DNA backbone**.
 - The **-10 region** is **also recognized by an α helix**. But in this case, the **interaction is less well-characterized** and is **more complicated because**:
 - The -35 region simply provides binding energy to secure polymerase to the promoter,
 - The **-10 region**, more elaborate role in transcription initiation, **within it DNA melting is initiated in the transition from the closed to open complex**.
 - Thus, the region of σ that interacts with the -10 region is doing more than simply binding DNA.
 - The **extended -10 element**, where present, is **recognized by an α helix in σ region 3**. This helix makes contact with the two specific base pairs that constitute that element.
 - **UP-element is not recognized by σ but is instead recognized by a carboxyl terminal domain (CTD) of the α subunit, called the α CTD.**
 - **The α CTD is connected to the α NTD by a flexible linker**. Thus, although the α NTD is **embedded in enzyme body**, the α CTD can reach the **upstream element** and can do so even when that element is not located immediately adjacent to the -35 region, but instead is located further upstream.
- σ and α subunits recruit RNA polymerase core enzyme to the promoter.
- The **position of α subunit** within the holoenzyme structure **makes recognition of various promoter elements feasible**. Thus, the **DNA-binding regions point away from the body of the enzyme** rather than being embedded, with the **spacing between those regions is consistent with the distance between the DNA elements they recognize**.
 - The σ regions 2 and 4 are separated by about 75 Å when σ is bound in the holoenzyme; and this is about the same distance as that between the centers of the -10 and -35 elements of a typical σ^{70} promoter.
 - Spacing of the protein domains is accommodated by the region 3 which is between α regions 2 and 4.
 - **Formation of the closed complex, is readily reversible**: polymerase can as easily dissociate from the promoter as make the transition to the open complex.

Step 2) Transition to the Open Complex Involves Structural Changes in RNA Polymerase and in the Promoter DNA:

- The σ -factor stimulates the polymerase to convert the closed promoter complex to an open promoter complex.
- The initial binding of RNA polymerase to the promoter DNA in the **closed complex** leaves the **DNA in double-stranded form**.
- The **next stage** in initiation **requires the enzyme to become more intimately engaged with the promoter**, in the open complex.
- The **transition**, often called **isomerization**, from closed to open complex involves **structural changes in the enzyme and the opening of the DNA double helix** to reveal the template and nontemplate strands. This "**melting**" occurs **between positions -11 and +3**, in relation to the transcription start site.
- In the case of the bacterial enzyme bearing σ^{70} , **isomerization**, does not require energy derived from ATP hydrolysis, and is instead the result of a spontaneous conformational change in the DNA-enzyme complex to a more energetically favorable form.
- **Isomerization is essentially irreversible** and, once complete, typically guarantees that transcription will subsequently initiate.
- Two striking structural changes are seen in the enzyme upon isomerization from the closed to open complex:

- 1) First, the **pincers** at the front of the enzyme **clamp down tightly on the downstream DNA**.
- 2) Second, there is a **major shift in the position of the N-terminal region of σ (region 1.1)**



- When **not bound to DNA**, σ **region 1.1** lies **within the active center cleft** of the holoenzyme, blocking the path that in the open complex, is followed by the template DNA strand.
- In the open complex, region 1.1 shifts some 50 Å and is now found on the outside of the enzyme, allowing the DNA access to the cleft.

- Region 1.1 of σ is highly negatively charged (just like DNA).
- Thus, **in the holoenzyme, region 1.1 acts as a molecular mimic of DNA.**
- The **space in the active center cleft**, which may be occupied either by region 1.1 or by DNA, is **highly positively charged**.

Transcription Is Initiated By RNA Polymerase Without The Need For A Primer:

- This **requires** that the initiating ribonucleotide be brought into the active site and held stably on the template while the next NTP is presented with correct geometry for the chemistry of polymerization to occur.
- It is **difficult because** RNA polymerase starts most transcripts with an A, and that ribonucleotide binds the template nucleotide (T) with only two hydrogen bonds (as opposed to the three between C and G).
- Thus, the **enzyme has to make specific interactions** with the initiating ribonucleotide, holding it rigidly in the correct orientation to allow chemical attack on the incoming NTP.
- The interactions are specific for that nucleotide (on A) and thus only chains beginning with A are held in a manner suitable for efficient initiation.
- The interactions are believed to be provided by various parts of polymerase holoenzyme, including part of sigma.

Step 3) Abortive Initiation followed by Formation of Stable Ternary Complex:

- Once **ribonucleotides enter the active center cleft** and **RNA synthesis begins**, there follows a period called **abortive initiation**.
 - In this phase, the enzyme synthesizes short RNA molecules of less than ten nucleotides in length.
 - Instead of being elongated further, these transcripts are released from the polymerase, and the enzyme, without disassociating from the template, begins RNA synthesis again.
- **It is not clear why RNA polymerase undergoes this period of abortive initiation.**
 - but once again a region of the σ factor, region 3.2, appears to be involved, acting as a molecular mimic and it mimics RNA.
 - This region of σ lies in the middle of the RNA exit channel in the open complex and for an RNA chain to be made longer than about ten nucleotides, this region of CT must be ejected from that location, a process that can take the enzyme several attempts.

- The ejection of σ region 3.2 probably accounts for σ being more weakly associated with the elongating enzyme than it is with the open complex; indeed it is often lost altogether from the elongating complex.
- Once a Polymerase manages to make an RNA longer than 10 bp, **a stable ternary complex** is formed.
- It is a complex containing:
 - the enzyme;
 - the DNA template;
 - and a growing RNA chain.
- This is the **start of the elongation phase**, which continues until polymerase is instructed to terminate transcription by specific sequences downstream of the gene.

Promoter Clearance and DNA Scrunching:

- It is the process by which an RNA polymerase moves away from a promoter after initiation of transcription.
- RNA polymerases cannot work if they do not recognize promoters, so they have evolved to recognize and bind strongly to them. But that poses a challenge when it comes time for promoter clearance: Somehow those strong bonds between polymerase and promoter must be broken in order for the polymerase to leave the promoter and enter the elongation phase.
- The **polymerase cannot move enough downstream to make a 10-nt transcript without doing one of three things**:
 1. moving briefly downstream and then snapping back to the starting position (**transient excursion**);
 2. stretching itself by leaving its trailing edge in place while moving its leading edge downstream (**inchworming**);
 3. or compressing the DNA without moving itself (**scrunching**).
- In 2006, Richard Ebright and colleagues showed that **scrunching appears to be the correct answer**.
- Ebright and colleagues examined both the leading and trailing edge of the *E. coli* RNA polymerase in complexes of polymerase attached to promoter DNA.
- Experiments showed that Polymerase does pull downstream DNA into itself (Scrunching).
- Thus, it may be the scrunched DNA that stores the energy expended in abortive transcript formation, rather like a spring, and enables the RNA polymerase finally to break away from the promoter and shift to the elongation phase.

- In another study, Ebright, Terence Strick, and colleagues used single-molecule DNA nanomanipulation to show that DNA scrunching indeed accompanies, and is probably required for, promoter clearance.
- **If scrunching occurs during abortive transcription**, then the **DNA will experience an extra unwinding**, which causes a compensating loss of negative supercoiling, or gain of positive supercoiling. Every unwinding of one helical turn (about 10 bp) leads to loss of one negative, or gain of one positive, supercoil. The change in supercoiling can be measured.
- **Strick found that unwinding of DNA accompanies formation of abortive transcripts**, and the degree of unwinding depends on the length of the abortive transcript made.
- To find whether scrunching also accompany promoter clearance, Ebright, Strick, and colleagues performed a study in which 80% percent of the transcription cycles studied had detectable scrunches. But 20% of the cycles were predicted to have scrunches that lasted less than 1 s, and 1 s was the limit of resolution in these experiments. So this 20% of cycles probably also had scrunches. The authors concluded that **approximately 100% of all the transcription cycles involve scrunching, which suggests that scrunching is required for promoter clearance.**

Reuse of σ factor:

Obligate Release Version of the σ -cycle model:

- In the same 1969 paper that showed sigma (σ) stimulates transcription initiation, **Travers and Burgess** demonstrated that σ can be recycled.
- Their experimental work supported the conclusion that σ gets released from the original core and can associate with a new core on a new DNA template.
- Accordingly, Travers and Burgess proposed that σ cycles from one core to another, They dubbed this the “ **σ cycle.**”
- The s-cycle model that arose from Travers and Burgess’s experiments called for the dissociation of σ from core as the polymerase undergoes promoter clearance and switches from initiation to elongation mode. This has come to be known as the **obligate release version of the s-cycle model.**

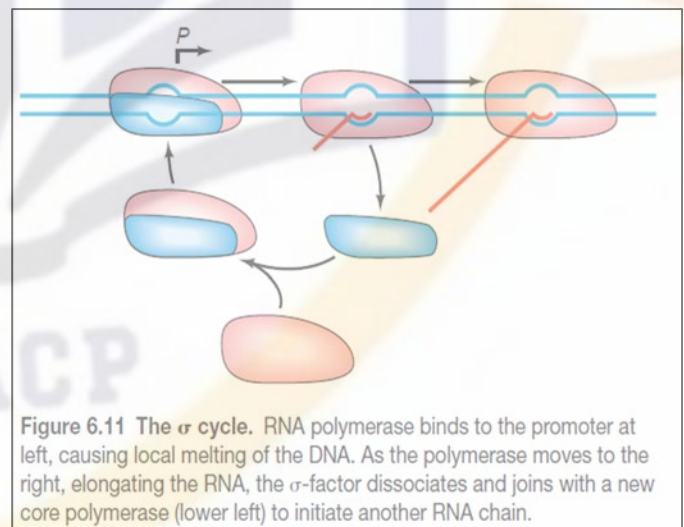


Figure 6.11 The σ cycle. RNA polymerase binds to the promoter at left, causing local melting of the DNA. As the polymerase moves to the right, elongating the RNA, the σ -factor dissociates and joins with a new core polymerase (lower left) to initiate another RNA chain.

The Stochastic σ -Cycle Model, an alternative view of σ cycle:

- Jeffrey **Roberts** and colleagues demonstrated, in **1996**, that σ is still attached to core polymerase at position +16/+17, well after promoter clearance has occurred.
- Based on this and other evidence, an alternative view of the σ -cycle was proposed: the **stochastic release model**.
- This hypothesis holds that σ is indeed released from the core polymerase, but there is no discrete point during transcription at which this release is required; rather, it is released randomly.
- Evidences in favor of this model were noted by **Ebright** and coworkers in **2001**, and by **Bar-Nahum** and **Nudler**, also in **2001**.
- Their results appear to rule out the obligate release model, and may seem to argue against the σ -cycle in general, they are actually consistent with the stochastic release version of the σ -cycle, which calls for σ release at multiple points throughout transcription.
- More studies by **Ebright** and coworkers confirmed that σ did indeed remain associated with the great majority (about 90%) of elongation complexes that had achieved promoter clearance (with transcripts 11 nt long).

Prepared by: M Anas Iftikhar, BS Zoology, PU